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Complexity measures: Characterization of transition to extreme events in dynamical systems

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ABSTRACT

In this work, we focus on different statistical measures to uncover the complexity of extreme events in various nonlinear dynamical systems. Notably, the effectiveness of the variance-based measure, estimated from the mean recurrence time and entropy, reveals a clear distinction between extreme and non-extreme events. Additionally, the variance measure clearly highlights the critical transition point of unforeseen, rare, large-amplitude dynamics. Our investigation involved models from various disciplines, including the Brusselator chemical oscillator, CO₂ laser, superconducting quantum interference devices, and a coupled neuron model. Deeper insights into the complexity of extreme events offer significant value for advancing early prediction methods.

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Over the past few decades, there has been a growing interest in exploring the complexity measures of various chaotic and stochastic dynamics across multiple disciplines. In recent times, uncovering the complexity of extreme events has received significant attention. Improving the comprehension of extreme events and their complex behaviors has important value for developing or improving the mitigation process against their catastrophic consequences. To progress on this pressing issue, we have used two distinct variance-based measures, namely, mean recurrence time and entropy, to reveal the complexity of extreme events. Our proposed measures identified the transition points of rare, large-amplitude dynamics and distinguished the regions of extreme and non-extreme events. The robustness of our approach has been examined in different classes of nonlinear dynamical systems.

I. INTRODUCTION

Understanding the catastrophic consequences of various extreme events, such as tsunamis, earthquakes, heatwaves, droughts,

forest fires, and stock-market crashes, is a primary issue in protecting nature and civilization. In recent decades, human life has been facing many challenges owing to destructive events such as various natural disasters, global warming, and economic crises, among others. Researchers in various domain have worked to uncover the emergence of extreme events in natural, empirical, and engineering systems.^{1–4} The ultimate goal of studying such catastrophic events is to develop an appropriate measure or tool for their earlier prediction. Identifying the complexity of distinct extreme events, as opposed to nominal chaos, is valuable for achieving the previously mentioned goal.

A large body of literature reported how to quantify the complexity of chaotic dynamics using Lyapunov exponents,⁵ Kaplan–Yorke dimensions,⁶ fractal dimensions,⁷ different entropy measures such as permutation, Kolmogorov–Sinai, Shannon, and others.^{8–11} Later, those investigations have been extended to distinguish noise from chaotic dynamics.¹⁰ On the other hand, the emergence of destructive phenomena, so-called extreme events, and their transition is explored in theoretical and experimental observations in various regions: hydrodynamics,^{12,13} optics and laser

models,^{14–16} neuronal models,¹⁷ electronic experiments^{18,19} superconducting devices,²⁰ and others (see Refs. 3 and 4 for more details). Importantly, the specific characteristic features of extreme events reveal that variations in their values within a range or oscillations occur in a confined region most of the time. However, there exist an occasional large excursions with shorter duration, i.e., appearance of higher magnitude events from their bounded region. The formation of rare and intense events has enhanced the complexity in their dynamics. As a consequence, the previously mentioned classical dynamical system analysis and statistical measures cannot identify the complexity of those rare and large-amplitude dynamics. Hence, it is an important issue to take into account when quantifying the complexity of such unusual dynamics for developing efficient forecasting tools in the future.

The recent studies disclosed that recurrence quantification analysis²¹ (RQA) has a significant advantage in exploring various instabilities, critical transition processes, regime shifts, and financial crashes, to name a few.^{22,23} In addition, there are a few reports that dealt with the advantages of RQA to understand mainly the dynamical evaluations of extreme events.^{24–26} However, in our illustration, we utilized the benefits of the variance measure in two different aspects: first, the variance is calculated from RQA analysis (i.e., mean recurrence time). Second, the variance of entropy is estimated from the temporal dynamics. Both measures quantify the complexity of extreme events and pinpoint their transition points.

The previous report on a time-delay-based gene regulatory network demonstrated the usefulness of the variance of mean recurrence time (MRT) to discriminate extreme and non-extreme events for the transitions of Pomeau–Manneville intermittency and quasiperiodic intermittency via unforeseen large-amplitude dynamics.²⁷ Nevertheless, some limitations persist in distinguishing from chaos to extreme event transition. Moreover, in some cases, extreme events and multiple large-amplitude oscillations exhibit similar maximal amplitudes; however, their dynamical processes and complexities differ significantly, and these dynamics have not yet been analyzed in detail.

In this study, we have utilized a simple and effective finite-time fluctuation measure, the variance of mean recurrence time and the variance of entropy, to exemplify the complexity of extreme events. In general, extreme events exhibit intermittent instability or nonstationary dynamics during their evolution. Similarly, the finite-time fluctuation measures exactly pinpoint such intermittent instability. Consequently, our approach correlates those two aspects and effectively captures the formation of rare large-amplitude events.

The outcome disclosed that the MRT variance significantly separated the bounded chaotic, multiple large-amplitude dynamics (also known as large-amplitude chaos), and extreme events, which originated through distinct dynamical processes. Furthermore, to confirm the effectiveness of the variance measure for the chaos to extreme events transition, we have analyzed the variance of entropy, which well matches the MRT variance estimation.

The recent report by Das *et al.*²⁸ illustrated that the combination of entropy and disequilibrium distance can be used as a complexity measure to discriminate extreme and non-extreme events in different model systems. Here, we propose a straightforward and effective variance-based measure that clearly reveals the intricate nature of extreme events and their transitions. Our earlier

works^{3,16,20,27} established that extreme events occur via three common routes: (i) Pomeau–Manneville intermittency, (ii) quasiperiodic intermittency, and (iii) interior-crisis-induced intermittency transition that is elaborated using a Zeeman laser,¹⁶ magnetically coupled superconducting devices,²⁰ and gene regulatory networks.²⁷ The primary focus of this work is to present the complexity measure of extreme events for those transitions using different classes of nonlinear dynamical systems. In particular, the transition from chaos to extreme events is delved into in detail using two different models, such as the Brusselator model²⁹ and a CO₂ laser.³⁰ The complexity measure of quasiperiodic intermittency and Pomeau–Manneville intermittency transitions is explored by utilizing the Superconducting QUantum Interference Devices (SQUIDS)²⁰ and a coupled Hindmarsh–Rose neuron model.^{31,32} A similar complexity measure is obtained for all transitions to extreme events regardless of the model.

The remaining part of the paper is organized as follows: In Sec. II, we analyze the details of the various complexity measures used for the investigations. In Sec. III, the complexity measures of the transition from chaos to extreme events are illustrated using two different classes of systems. The transition from quasiperiodic or periodic to unforeseen large-amplitude dynamics and their relevant complexity analysis are elaborated in Sec. IV. The key observation of our findings is discussed in the final section.

II. COMPLEXITY MEASURES

In this section, we explain the details of statistical measures that are used to analyze the complexity of extreme events. In the first step, the time series is partitioned into k non-overlapping segments of fixed duration T . Then, $x^{(j)} = \{x_{(j-1)T+1}, \dots, x_{jT}\}$, $j = 1, 2, \dots, k$ denotes the finite segments corresponding to the j th window. For each segment, the sequence of local maxima is given by $E^{(j)} = \{e_i^{(j)}\}_{i=1}^{n_j}$, where n_j is the number of peaks in the j th segment. Furthermore, we utilize the peak sequence of each segment to capture the finite-time dynamical process.

A. Mean recurrence time

A recurrence plot (RP) is a binary matrix representation that manifests when a state of a time series reappears, i.e., when it returns to a neighborhood of a previous state within a predefined threshold. Indeed, for a univariate discrete time series $x[n]$, the recurrence matrix is defined as

$$R[i, j] = \begin{cases} 1 & \text{if } d(x[i], x[j]) \leq \epsilon, \\ 0 & \text{otherwise,} \end{cases}$$

where d is a distance metric and ϵ is the recurrence threshold. Each axis of the RP corresponds to the same time series, and the binary entries highlight pairs of times where the system revisits similar states. This makes RPs a powerful tool for visualizing periodicity, patterns, and other repetitions within time series data.²¹

The selection of the recurrence threshold ϵ is essential in RP analysis. Although numerous studies have addressed threshold selection,^{33,34} the universally accepted value is still in debate. For classification tasks and signal detection, a reasonable suggestion is to set the ϵ within 20%–40% of the signals' standard deviation σ

(see Ref. 33). In our investigations, we utilized 30% and employed the Euclidean distance norm (L_2 -norm). The recurrence matrix is a sparse square matrix of Boolean values that quantifies recurrences in the trajectory, indicating locations where the system returns to a previous state.

Next, for given temporal data, the mean recurrence time (MRT) is determined by calculating the average of the frequency distribution $p(l)$ of white vertical lines of length l .³⁵ The mathematical representation of MRT is described as follows:

$$\text{MRT} = \frac{\sum_{l=1}^N lp(l)}{\sum_{l=1}^N p(l)}. \quad (1)$$

Here, the peak sequence of each segment $E^{(j)}$ is used for the MRT estimation. When the system is in the bounded chaotic motion, the mean recurrence time remains closely consistent or exhibits minor variations during the entire time evolution. On the other hand, the segments that contain extreme events show higher MRT values as compared with the MRT values of non-extreme events' segments.

B. Entropy

There exist distinct types of entropy (ENT) in the literature to characterize numerous intricate dynamics.⁹ Here, we have used Shannon entropy H to quantify the complexity of extreme events and other dynamics. The Shannon entropy H is computed based on the peak sequence data that is described as follows:³⁶

$$H = - \sum_{i=1}^m p_i \log_2 p_i, \quad (2)$$

where p_i denotes the probability of values falling into the i th bin. For each data sequence $E^{(j)}$, the values are divided into m equal-width bins, where the bin width is given by $(\max E^{(j)} - \min E^{(j)})/m$. Then, the normalized Shannon entropy is calculated based on the following notation:

$$\text{ENT} = H/H_{\max}, \quad (3)$$

where $H_{\max} = \log_2 m$. The zero or minimal ENT values represent regular dynamics. In contrast, the gradual increase of higher values of ENT showed the existence of a disordered state in the system.

C. Variance

As mentioned earlier, in our investigation, the variance-based measure was used to unveil the complexity of different intricate dynamics, including extreme events. The variance is estimated using squared deviations from their respective mean values. Suppose the data that has been used for evaluation has a few large deviated values from its mean range; then the system manifests high variance.

Then, the variance of MRT values of the k segments is calculated using the following expression:

$$V_{\text{MRT}} = \frac{1}{k-1} \sum_{j=1}^k [\text{MRT}(E^{(j)}) - \overline{\text{MRT}}]^2, \quad (4)$$

where $\text{MRT}(E^{(j)})$ is the mean recurrence time value for the segment j , and $\overline{\text{MRT}}$ denotes its mean value. Similarly, the variance of entropy (V_{ENT}) is also estimated.

III. CHAOS TO EXTREME EVENT TRANSITION

This work focuses on leveraging a specific RQA and a statistical measure to identify the critical transition points of extreme events and their changeover in various nonlinear dynamical systems. The interpretation begins by examining the complicated transition of chaos to extreme events.

A. Brusselator model: Period-doubling routes to the chaos region

First, we addressed the natural question of how the values of MRT and ENT differ within the distinct periodic and bounded chaotic region. To present that, the above-stated measures are estimated for the chemical system of the Brusselator model.^{29,37} The system is represented by the following differential equations:

$$\begin{aligned} \frac{du_1}{dt} &= \mu - (\alpha + 1)u_1 + u_1^2 u_2 + I \sin(\omega t), \\ \frac{du_2}{dt} &= \alpha u_1 - u_1^2 u_2. \end{aligned} \quad (5)$$

The normalized system parameters in Eq. (5), such as μ and α , denote the primary concentration of the reactant and the decaying rate of the auto-catalyst, respectively. Additionally, an external forcing is incorporated into the model to illustrate the emergence of various instabilities in the system. The parameter I represents the amplitude of the external forcing, and ω signifies its respective frequency. For the fixed parameter values of $\mu = 0.2$, $\alpha = 1.114$, $\omega = 0.69$, and a gradual increase of I , the Brusselator model elucidated successive period-doubling routes to chaos. We have presented the bifurcation diagram of period-doubling routes to chaos for a range of $I \in (0.051, 0.0518)$ in Fig. 1(a). The color bar in the bifurcation diagram describes the number of occurrences of different dynamics for a specific length of time series. The bifurcation diagram disclosed the frequent appearances of periodic dynamics (dark green). However, the system exhibits considerable variations in the occurrences of bounded chaotic motion (ranging from light green to yellow) across different amplitude levels, from lower to higher [see Fig. 1(a)].

Then, we have calculated the MRT values for the same range of bifurcation parameters that is pictured as a blue line in Fig. 1(b). The MRT measure exhibits different constant values for the distinct periodic attractors and begins to fluctuate as the system enters the chaotic region. Specifically, for the range of bounded chaotic regions, MRT values are small as compared with different periodic dynamics delineated in Fig. 1(b). As a consequence, to identify the clear distinction between regular and chaotic dynamics, we have analyzed the variance of MRT values (V_{MRT}) and their outcome illustrated in Fig. 1(b) (red line). The Brusselator model exhibits null variance for the periodic dynamics and shows small variance for the chaotic region, which is visible in Fig. 1(b).

Subsequently, we have estimated the entropy values for the period-doubling routes to the chaos region, which is presented as a blue line in Fig. 1(c). The system shows minimal entropy values for different periodic dynamics and a gradual increase of entropy values for the chaotic region. Likewise, the variance of entropy (V_{ENT}) is calculated, and their results are presented as a red line in Fig. 1(c).

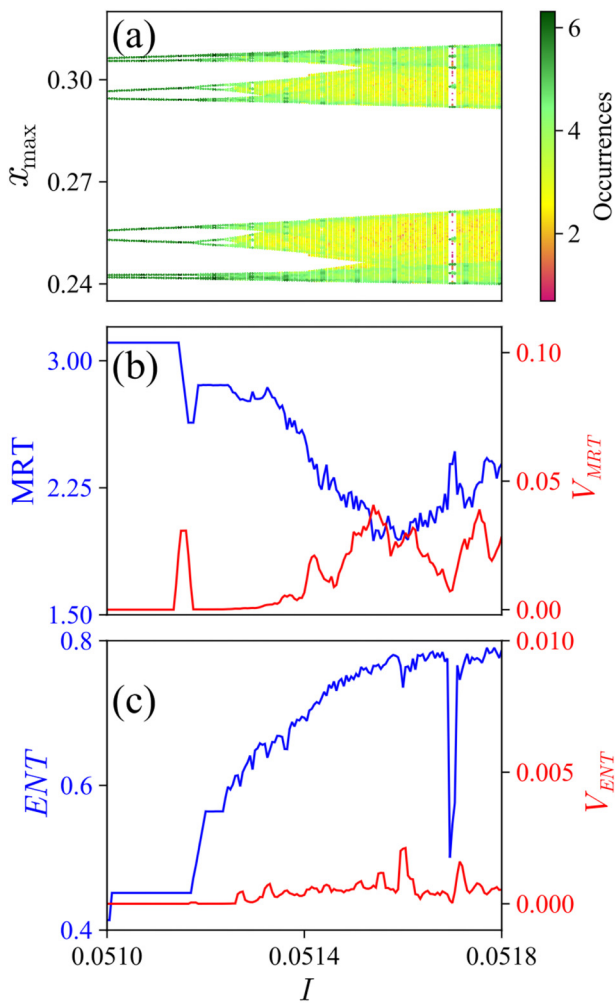


FIG. 1. Period-doubling routes to chaos: (a) one-parameter bifurcation for the range of $I \in (0.051, 0.0518)$ delineates the consecutive occurrence of different periodic states and a range of variations in their oscillations for chaotic dynamics using a fixed length of time series data. (b) Mean recurrence time (MRT) and entropy (ENT) measures, and their corresponding variances, such as V_{MRT} and V_{ENT} , signify different complexities among periodic and bounded chaotic dynamics.

The Brusselator exhibits a zero V_{ENT} value for the regular dynamics and a smaller swing in its values for the region of the chaotic motion. On the other hand, the system demonstrates notable changes in its V_{MRT} and V_{ENT} values, while the chemical model that manifests occasional large deviations from the bounded chaotic dynamics will be discussed in the forthcoming section.

B. Brusselator model: Chaos to extreme event transition

Now, we turn our attention to how to distinguish rare large-intensity events (extremes) from other dynamics in the

Brusselator model. To interpret that the control parameter I of Eq. (5) is increased $I > 0.0519$ (i.e., beyond the chaotic region) and the system manifests rare large-amplitude events from the bounded chaotic states. The abrupt large-amplitude dynamics appeared at the critical parameter value due to the mechanism of interior-crisis-induced intermittency.³ This process refers to the fact that the system evolves in bounded chaotic motion most of the time, and there exist occasional bursts while the chaotic attractor collides with unstable periodic orbits. Subsequently, the trajectory returns to its bounded regions due to the stretching and folding nature of chaotic motion. As a consequence, the system shows an enormous increase in its amplitude at the transition point during this process. The transition from chaos to extreme events is presented as a one-parameter bifurcation diagram in Fig. 2(a). The bifurcation diagram portrayed in Fig. 2(a) manifests the appearance of three different dynamics in the system: (i) bounded chaotic dynamics for $I \in (0.051, 0.0519)$, (ii) extreme events for $I \in (0.0520, 0.0545)$, and (iii) frequent large-amplitude dynamics for $I \in (0.0546, 0.056)$. The different dynamics of extreme and non-extreme (bounded chaos and multiple large-amplitude dynamics) events were distinguished using a well-known extreme events qualifier threshold (Q_{th}). Here, the value of Q_{th} is defined as the sum of the mean plus six times the standard deviation of the local maxima ($x_{\max}$) of the temporal evolution ($Q_{th} = \langle x_{\max} \rangle + 6\sigma$; where σ denotes the standard deviation). The obtained Q_{th} threshold is plotted against the bifurcation diagram as a black line in Fig. 2(a), which differentiates the appearance of extreme and non-extreme event regions in the system. The color bar in the bifurcation diagram shows a recurrent appearance of bounded chaotic dynamics (indicated by the dark to light green region) and a rare occurrence of maximum amplitude (extremes) represented by the dark yellow to dark pink region across a range of control parameter values. Once again, when I reaches into the region of frequent large-amplitude events (also known as large-amplitude chaos), the number of occurrences increased owing to the recurrent appearance of large-amplitude motions. Accordingly, the number of occurrences varies from light yellow to light green range is clearly visible in Fig. 2(a). Note that in Fig. 2(a), in the range of $I (0.0514, 0.0519)$, the system shows chaotic oscillation [see Fig. 1(a) for clarity]. The temporal dynamics of extreme events for the Brusselator model and its respective distribution are presented in the Appendix.

The MRT values are calculated for the chaos to extreme event transition and its result is presented as a blue line in Fig. 2(b). The system exhibited lower MRT values for the chaotic region and started increasing when it switched to the extreme events regime. Then, for the multiple large-amplitude dynamic regions, the system shows moderately higher MRT values. Despite that, a closer inspection of the MRT measure disclosed that there exists an uncertainty in their results; that is, MRT values are the same or nearly close for both extreme and multiple large-amplitude dynamics. On the contrary, the dynamical process including the complexities are different for extreme events and large-amplitude chaotic motion. Hence, to establish a precise difference between those two distinct dynamics, we have calculated the variance of mean recurrence time (V_{MRT}), and its result is pictured as a red line in Fig. 2(b). Intriguingly, V_{MRT} measure shows zero or smaller variance for the non-extreme events' dynamics (chaos and frequent large-amplitude motion) and higher variance for the extreme events' region as shown in Fig. 2(b).

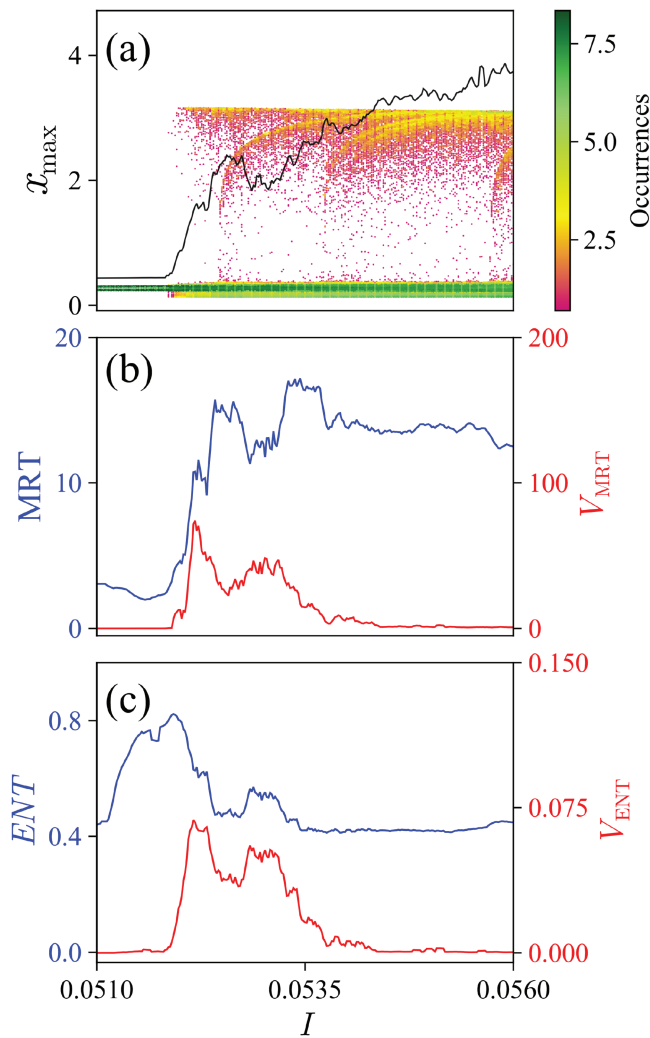


FIG. 2. (a) The bifurcation diagram shows the transition from chaos to extreme events in a Brusselator model. The minimal or rare occurrence of large-amplitude events in the color bar signifies extreme events, and the recurrent existence of large-amplitude oscillations, including bounded chaotic motion, shows a larger number of occurrences. The number of occurrences in the bifurcation is denoted on a log scale for better visualization. The extreme events' qualifier threshold (Q_{th}) is plotted as a black line in (a). The MRT and ENT values are signified as blue lines in (b) and (c), and their respective variances are showcased as red lines [V_{MRT} (b) and V_{ENT} (c)]. This exhibited a clear distinction between extreme and non-extreme events.

In view of ensuring the applicability of the variance-based measure, we have calculated the ENT and variance of entropy (V_{ENT}) for the same range of bifurcation parameter region delineated as blue and red lines in Fig. 2(c). The simple ENT analysis failed to detect some part of the extreme events region. In contrast, the V_{ENT} measure provided the precise difference between extreme and other dynamics. For instance, chaotic and multiple large-amplitude events, we have obtained minimum V_{ENT} values, whereas higher

V_{ENT} values were captured for the extreme events region. Remarkably, both variance measures identified the critical transition point of the system, respectively. Moreover, our study eliminates the limitation of exploring the extreme events and large-amplitude chaos reported in Ref. 28. As a consequence, the more profound insight into the variance-based measure sheds light on exploring the complexity of extreme events.

In addition, we have performed a similar analysis for an extended range of parameter space by presenting phase diagrams in Fig. 3. The formation of extreme and non-extreme events' region is shown in Fig. 3(a), for a range of $I \in (0.05, 0.07)$ and $\omega \in (0.66, 0.72)$. A specific quantitative measure *count* has been used to differentiate extreme and other events in the system. The count measure indicates that, for a sufficiently long temporal dataset, rare large-amplitude events that exceed the qualifier threshold are calculated as count; the mathematical description of this measure is discussed in Ref. 16. The light pink region denotes the existence of non-extreme events in the chemical model, and the variation of the dark pink to dark green region represents the counts of rare large-amplitude events in the system. The count of extreme events varies with the influence of system parameters. We have used a log scale in the color bar to showcase the number of extreme events (counts) for the purpose of better visualization. Furthermore, we have computed the variance of MRT and variance of ENT for a similar range of phase diagrams, and their respective results are shown in Figs. 3(b) and 3(c). The system showed higher variance in the extreme events region and lower variance in the periodic, chaotic, and multiple large-amplitude dynamics. The existence region of extreme events in Fig. 3(a) well matches the MRT variance and ENT variance phase diagrams [see Figs. 3(b) and 3(c)]. To establish the robustness of our observation, a similar chaos to extreme event transition will be presented in Sec. III C, using a different class of non-smooth dynamical system.

C. CO₂ laser model

The emergence of extreme events has been reported in a non-smooth dynamical system of optically modulated CO₂ laser.³⁰ Its governing equations are read as follows:

$$\begin{aligned} \dot{x} &= x [I_1 (y - 1.0 - k_1 \sin^2 z) - I_3 \epsilon \cos(I_2 t)], \\ \dot{y} &= -(y - n) - 2I_1 xy, \\ \dot{z} &= -I_4 \left(z - b + \frac{rx}{1 + \zeta x} \right). \end{aligned} \quad (6)$$

The details of the normalized model equation and its relevant information are given in Refs. 30 and 38. Compared to the Brusselator chemical model, the CO₂ laser also exhibits rare large-amplitude dynamics from the bounded chaotic motion. However, the dynamical processes involved during the transition to extreme events are different. There exists a discontinuous boundary in the CO₂ laser at $x = 0.0$; the system revealed various periodic and chaotic dynamics until the trajectories approached the discontinuous boundary. At a critical control parameter value, the systems' trajectories reached into the vicinity of the discontinuous boundary, a phenomenon known as grazing dynamics. Afterward, the trajectory becomes

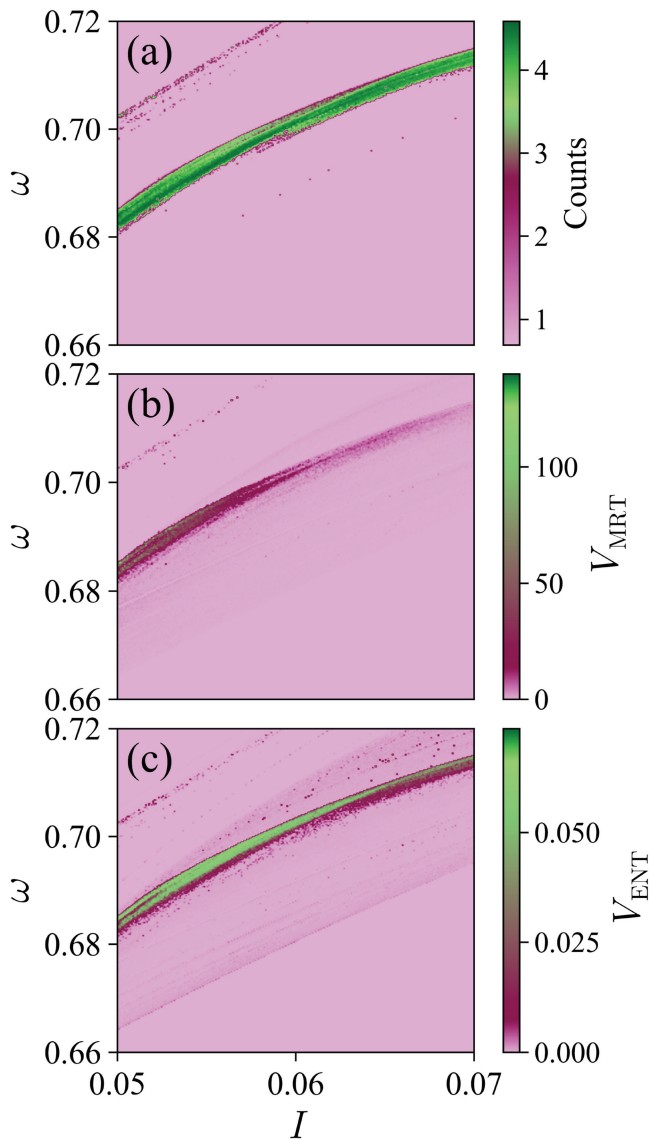


FIG. 3. (a) The phase diagram of a chemical oscillator shows the emergence of extreme (dark pink to dark green) and non-extreme (light pink) events in a parameter space. Its equivalent V_{MRT} (b) and V_{ENT} (c) measures signify higher variance in the region of extreme events.

trapped and slides along the area near the discontinuous boundary. Hence, large deviation appeared in the system owing to grazing-sliding bifurcation. Note that those grazing-sliding dynamics manifested in the CO₂ laser at irregular time durations, while the trajectories reached closer to the discontinuous boundary. Additional details about this process are discussed in Ref. 30. The system parameters are fixed as follows: $n = 4.0$, $b = 4.5$, $\eta = 25.57$, $l_1 = 28.57$, $l_2 = 30.0$, $l_3 = 25.0$, $l_4 = 6.452$, and $r = 200$. By changing ϵ in the range (1.625, 1.685), the CO₂ laser showed bounded chaotic dynamics followed by extreme events and then transformed

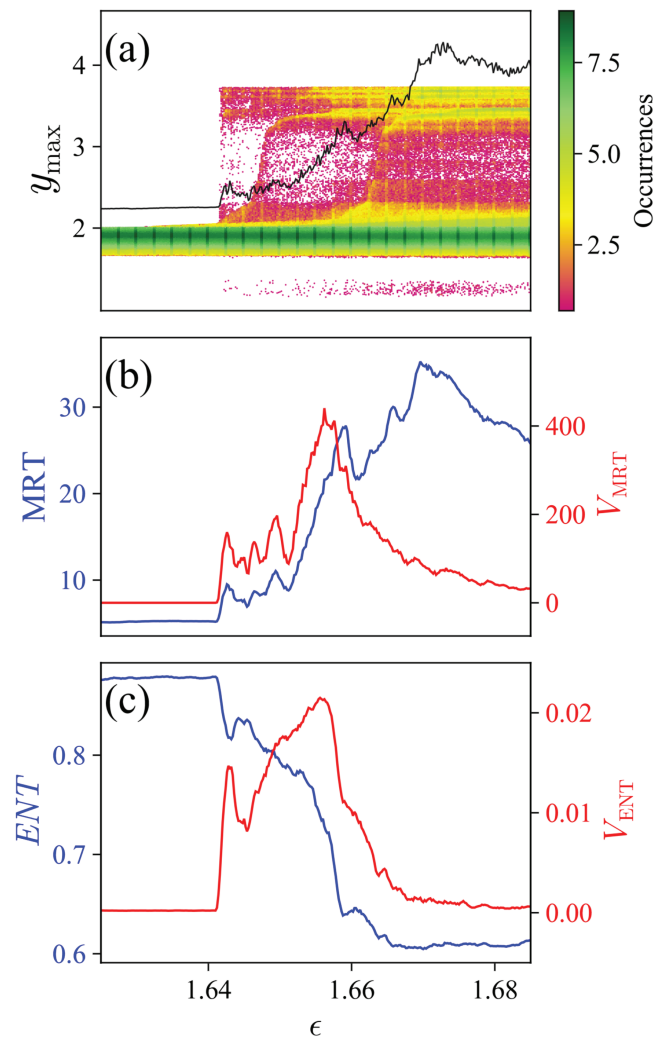


FIG. 4. Chaos to extreme event transition in a non-smooth dynamical system: (a) Bifurcation diagram shows the formation of extreme events after the chaotic dynamics for the specific range of ϵ . The extreme events' qualifier threshold is plotted as a black line in (a). Blue lines in (b) and (c) signify the MRT and ENT measures of non-extreme and extreme events, with their respective variance plotted as red lines in (b) and (c).

into multiple large-intensity events. The transition from chaos to extreme events is delineated as a bifurcation diagram in Fig. 4(a). The color bar in the bifurcation plot clearly represents the recurrent appearance of chaotic and multiple large-intensity dynamics and the rare existence of extreme events in the CO₂ laser. The Q_{th} value of $\langle y_{max} \rangle + 6\sigma$ is plotted over the bifurcation diagram to confirm the existing region of extreme events in the system. Time evolution of extreme events and their distribution is showcased in the Appendix.

The MRT values are calculated for both non-extreme and extreme events, inclusive of their corresponding variance (V_{MRT}), which is shown in Fig. 4(b). The MRT shows a moderately higher

magnitude for the chaotic region and starts increasing in its values while the system swaps from chaos to extreme events. However, the MRT values for the region of multiple large-intensity dynamics are gradually decreasing, as depicted in Fig. 4(b). Additionally, the estimation of the variance of MRT (V_{MRT}) distinguishes between non-extreme and extreme events; specifically, minimal V_{MRT} values are observed for both bounded chaotic and frequent large-intensity dynamics, while higher V_{MRT} magnitudes are evident for extreme events, as shown in Fig. 4(b). Next, the entropy and variance of ENT

are calculated for a similar transition, and their results are demonstrated in Fig. 4(c). For the chaotic region, ENT values are sustained near one, and their magnitude begins decreasing for the extreme events region. Subsequently, for the frequent large-intensity dynamics, the ENT value remains at a minimal level. On the other hand, the variance of entropy (V_{ENT}) provides crucial information, indicating that higher values correspond to extreme events while lower values are associated with non-extreme regions, as shown in Fig. 4(c).

In addition, we have presented a phase diagram for the CO₂ laser by varying $\epsilon \in (1.62, 1.66)$ and another system parameter r in a range $r \in (199, 201)$ depicted in Fig. 5(a). The light pink region denoted the region of non-extreme events, whereas the variation of count from the dark pink to the dark green region showed the number of occurrences of extreme events (counts) in the CO₂ laser. We calculated the variance of MRT and ENT for the same parameter space, and the results are in Figs. 5(b) and 5(c). V_{MRT} and V_{ENT} exhibited minimal variance for the non-extreme events and higher variance (dark pink to dark green) for the extreme events' regions. The color bar of Fig. 5(a) is presented in a log scale for the better interpretation of the results.

IV. PERIODIC AND QUASIPERIODIC INTERMITTENCY TRANSITION TO EXTREME EVENTS

In the earlier section, we presented the more complex transition of chaos to extreme events. Now, we focus on the other two leading transitions to extreme events: (i) Pomeau–Manneville and (ii) quasiperiodic intermittencies in different models.

A. SQUID trimer

To illuminate the quasiperiodic intermittency transition, we have taken the three magnetically coupled Superconducting QUantum Interference Devices (SQUIDs).³⁹ The normalized form of the model equation is represented by the following differential equations:

$$\begin{bmatrix} \ddot{\phi}_1 + \gamma \dot{\phi}_1 + \beta \sin(2\pi \phi_1) \\ \ddot{\phi}_2 + \gamma \dot{\phi}_2 + \beta \sin(2\pi \phi_2) \\ \ddot{\phi}_3 + \gamma \dot{\phi}_3 + \beta \sin(2\pi \phi_3) \end{bmatrix} = \frac{1}{D} (F) \begin{bmatrix} \phi_1 - \phi_{\text{ext}} \\ \phi_2 - \phi_{\text{ext}} \\ \phi_3 - \phi_{\text{ext}} \end{bmatrix}. \quad (7)$$

Here, F is denoted as

$$F = \begin{bmatrix} 1 - \Lambda^2 & -\Lambda + \frac{\Lambda^2}{8} & \Lambda^2 - \frac{\Lambda}{8} \\ -\Lambda + \frac{\Lambda^2}{8} & 1 - \frac{\Lambda^2}{64} & -\Lambda + \frac{\Lambda^2}{8} \\ \Lambda^2 - \frac{\Lambda}{8} & -\Lambda + \frac{\Lambda^2}{8} & 1 - \Lambda^2 \end{bmatrix},$$

and $D = \det(\hat{\Lambda}) = (1 - \frac{129}{64}\Lambda^2 + \frac{1}{4}\Lambda^3)$.

An additional driving force is incorporated in the circuit, which is described by $\phi_{\text{ext}} = \phi_{\text{dc}} + \phi_{\text{ac}} \cos(\omega\tau)$. The parameter Λ represents the magnetic coupling strength among the SQUIDs. The remaining normalized parameters are expressed as $\beta = \frac{I_c L}{\Phi_0}$

and $\gamma = \frac{1}{R} \sqrt{\frac{L}{C}}$. Also, Φ_0 denotes the magnetic flux quantum constant: $\Phi_0 = 2.067833 \times 10^{-15}$ Wb. The value of β is chosen as $\beta = 0.1384$, and ϕ_{dc} is considered zero for our illumination. Then,

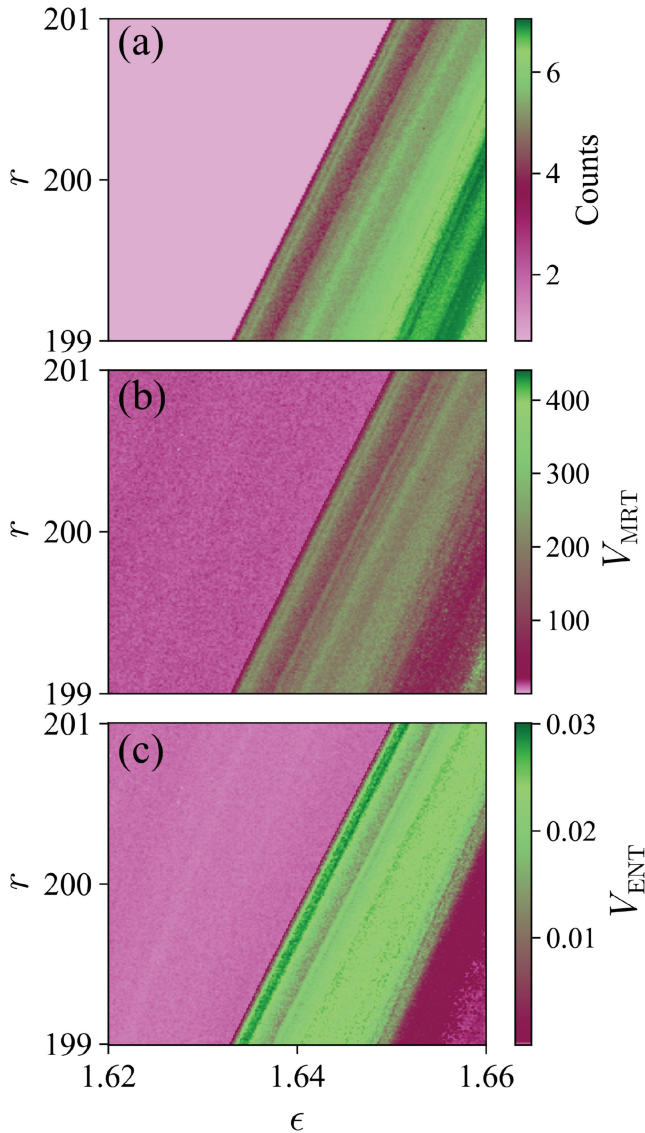


FIG. 5. (a) Phase diagram of the CO₂ laser model, which describes the regions of extreme and non-extreme events. (b) The variance of MRT and (c) variance of ENT exhibit minimal variance for non-extreme events (light pink) and higher variance for the extreme events' region (dark pink to dark green), respectively.

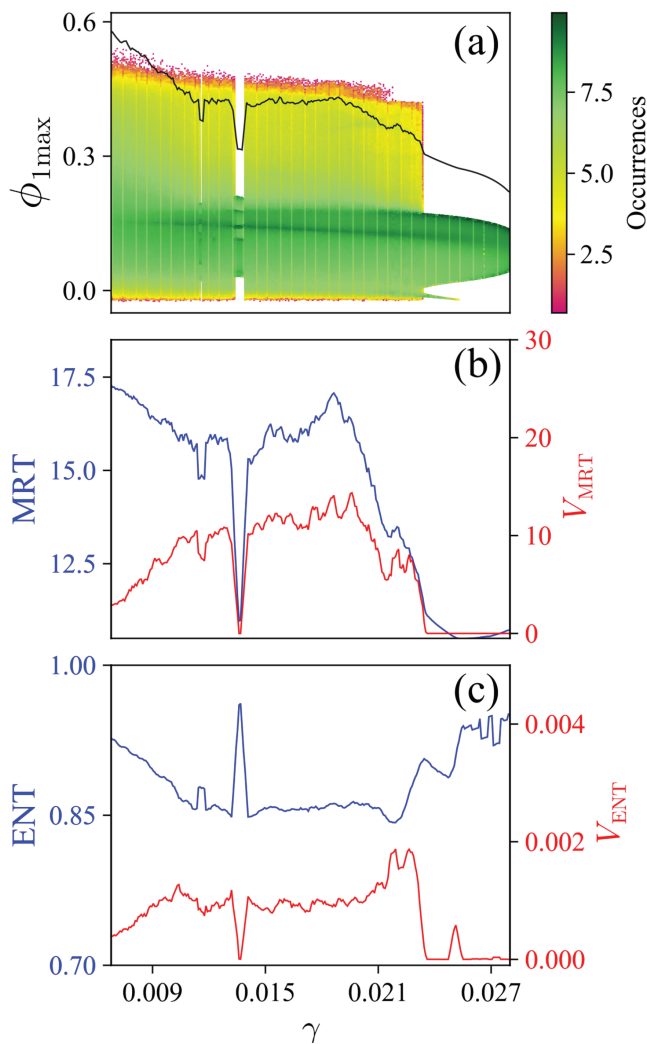


FIG. 6. (a) Bifurcation diagram of SQUID trimer manifests quasiperiodic intermittency transition to extreme events. The blue lines in (b) and (c) signify MRT and ENT measures; their corresponding variances in red lines (V_{MRT} and V_{ENT}) clearly differentiate between the regions of extreme and non-extreme events.

for the fine-tuning of the control parameter γ , the SQUID trimer shows quasiperiodic intermittency transition via occasional large-amplitude dynamics.²⁰ The advent of quasiperiodic intermittency is a unique phenomenon that has been reported only in a limited number of nonlinear systems.^{27,40,41} Quasiperiodic motion in its laminar phase exhibits rare bursting (turbulent phase) at irregular time intervals at a critical value of control parameters; this transformation process is known as quasiperiodic intermittency. Here, the SQUID trimer signifies rare bursts from the confined quasiperiodic region that also exceed the extreme events' qualifier threshold value.

One-parameter bifurcation diagram for the SQUID trimer presented in Fig. 6(a) by varying γ in the range $\gamma \in (0.006, 0.028)$. The quasiperiodic motion originated after the period dynamics in

the system due to Neimark–Sacker bifurcation (not shown here; for details, see Ref. 20). The quasiperiodic dynamics exist in the system for the range of $\gamma \in (0.028, 0.0235)$. At the critical control parameter value, say, $\gamma = 0.0234$, the SQUID trimer proves occasional bursting from quasiperiodic motion at infrequent time intervals; such dynamics, known as quasiperiodic intermittency, have been rarely identified in a broader range of nonlinear dynamical systems.⁴² The number of occurrences in the bifurcation diagram shows the frequent existence of quasiperiodic motion and the rare appearance of extreme events in the SQUID model. The color bar is presented in a log scale. We have computed the Q_{th} values and superimposed them with the bifurcation diagram as a black line shown in Fig. 6(a) to locate the extreme events' region. For computing the qualifier threshold estimation, we have used the mean plus four times the standard deviation of local maxima of the temporal dynamics ($Q_{th} = \langle \phi_{1\max} \rangle + 4\sigma$).

Moreover, the MRT and ENT values are calculated for quasiperiodic, quasiperiodic intermittency, and large-amplitude chaotic motions, presented as blue lines in Figs. 6(b) and 6(c). The MRT and ENT signify lower and higher values for quasiperiodic motion and attain a minimal magnitude at the transition point. Additionally, the MRT value begins to increase for the quasiperiodic intermittency dynamics, while the ENT measures exhibit a range of values, as shown in Figs. 6(b) and 6(c). On the other hand, when calculating the variance of MRT and ENT, it is evident that the quasiperiodic region exhibits minimal variance, while the quasiperiodic intermittency region displays higher variance (red lines) in Figs. 6(b) and 6(c). Furthermore, when the quasiperiodic intermittency motion (extreme events) turns into a large-amplitude chaotic oscillation, their respective variance of MRT and variance of ENT values decay due to the frequent appearance of large-amplitude oscillations.

B. Hindmarsh–Rose neuron model

Finally, to illustrate the Pomeau–Manneville intermittency via extreme bursting dynamics, we have analyzed a coupled Hindmarsh–Rose neuron model.^{3,32} The neuron model is described by the following differential equations:

$$\begin{aligned} \dot{x}_i &= y_i + bx_i^2 - ax_i^3 - z_i + I - k_i(x_i - v_s)\Gamma(x_j), \\ \dot{y}_i &= c - dx_i^2 - y_i, \\ \dot{z}_i &= r[s(x_i - x_R) - z_i]. \end{aligned} \quad (8)$$

Here $i, j = 1, 2$ ($i \neq j$) signify two coupled neurons. The interaction between them is activated through the chemical synaptic coupling that is described as a sigmoidal function $\Gamma(x) = \frac{1}{1 + e^{-\lambda(x - \Theta)}}$. The parameter values of the coupling function are set as $v_s = 2.0$, $\lambda = 10.0$, and $\Theta = -0.25$. Similarly, other system parameters are chosen as $a = 1.0$, $b = 3.0$, $c = 1.0$, $d = 5.0$, and $x_R = -1.6$. The individual neuron shows periodic bursting dynamics before activating the coupling function. For the suitable choice of the coupling strength, the HR model shows the PM intermittency transition via extreme events. The changeover in the system dynamics from a laminar phase of periodic dynamics to a turbulent phase with the influence of a system parameter, so-called Pomeau–Manneville intermittency.^{42,43} In our investigation, during Pomeau–Manneville

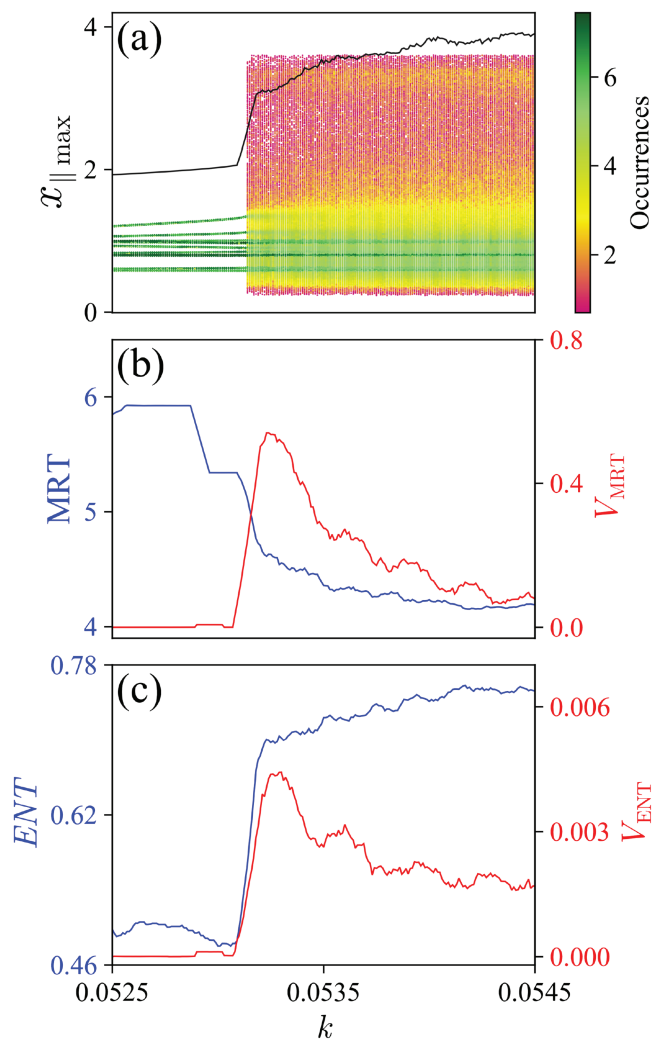


FIG. 7. (a) The one-parameter bifurcation diagram of the coupled neuron model shows Pomeau–Manneville intermittency via extreme bursting dynamics. The MRT and ENT measures (blue lines) in (b) and (c) have some limitations in distinguishing extreme events and other dynamics. However, the variance-based measures of V_{MRT} and V_{ENT} (red lines) in (b) and (c) illustrated the precise difference between extreme and non-extreme events.

intermittency, the system reveals occasional large-amplitude bursting (turbulent state) over a threshold (Q_{th} value) from the periodic motion. We have specifically taken $k_1 = k_2 = k$, which is a positive value that denotes the excitatory or attractive coupling between the neurons.

The bifurcation plot of the HR model, for the range of coupling strength $k \in (0.0525, 0.0545)$, is shown in Fig. 7(a). The extreme event qualifier threshold $Q_{th} = \langle x_{\parallel} \rangle + 6\sigma$ is plotted as a black line in the bifurcation diagram of Fig. 7(a). Here, $x_{\parallel}$ denotes $x_{\parallel} = x_1 + x_2$, and we have used the $x_{\parallel}$ variable to perform all the computation for this model. For the periodic dynamics, Q_{th} is placed far from its

maximum amplitude, and extreme events cross the Q_{th} threshold value. The HR model exhibited Pomeau–Manneville intermittency dynamics at the critical coupling value $k = 0.0532$. At this point, the neurons show rare bursts from the periodic region at sporadic time durations. The temporal dynamics of Pomeau–Manneville intermittency originated extreme events and their associated probability distribution, given in the Appendix.

The MRT and ENT values for the periodic, Pomeau–Manneville intermittency, and frequent large-amplitude bursting dynamics are portrayed as blue lines in Figs. 7(b) and 7(c). The MRT values show a higher magnitude for the different periodic attractors; it begins decreasing when their dynamics switch into extreme events. Furthermore, its magnitude saturated for the large-amplitude chaotic bursting [see Fig. 7(b)]. The ENT measure shown in Fig. 7(c) exhibits gradual increases in their values for the periodic region and swaps in a range of values for the rare large-amplitude bursting region. Subsequently, there is a slight increase in the values for the large-amplitude chaotic bursting. Accordingly, we have calculated the variance-based measures of V_{MRT} and V_{ENT} that are pictured as red lines in Figs. 7(b) and 7(c). Both variance measures manifest zero (minimal) variance for the periodic (multiple large-amplitude bursting) region and higher variance for extreme bursting oscillation.

V. CONCLUSION

We have explored distinct variance-based measures for identifying the complexity of extreme events, among other dynamics. The mean recurrence time approach, based on recurrence quantification analysis, has some limitations in elucidating the complexity of chaos and extreme events. However, the variance of the mean recurrence time analysis disclosed a precise difference between the chaos and extreme events. To confirm the robustness of the variance measure, we have computed the variance of entropy using its temporal dynamics. The obtained results show a clear distinction between the extreme and non-extreme events of recurrent large-amplitude dynamics, bounded chaos, quasiperiodic motion, and distinct periodic states.

Our investigation involved analyzing the complexity measure of three major pathways to extreme events in different models. To analyze the transition from chaos to extreme events, we have examined two different systems: the continuous-time dynamical system of the Brusselator model and the non-smooth dynamical system of a CO₂ laser. However, the dynamical origins for the appearance of extreme events are different. In the chemical model, extreme events appeared via interior-crisis-induced intermittency, and the CO₂ laser exhibits extreme pulses owing to grazing-sliding bifurcation. In addition, we have analyzed the complexity measure of quasiperiodic intermittency and Pomeau–Manneville intermittency-induced extreme events using a SQUID trimer and a coupled Hindmarsh–Rose neuron model. Notably, we have utilized wider classes of nonlinear systems to reveal the complexity of extreme events and other dynamics that arise from various dynamical mechanisms. The outcome shows a substantial increase in the variance measure for all the examples. Hence, the proposed indicator is model-independent and remains consistent across various emerging processes.

The bifurcation diagrams are used to represent the rare occurrence of extreme events, the range of variation in their amplitude for chaotic dynamics, and the consecutive existence of periodic/quasiperiodic motion during the transition to extreme events. The temporal dynamics and probability distributions confirm the advents of extreme events in different systems. The comprehensive overview of the complexity of extreme and other events would provide a new avenue for improving the forecasting techniques against distinct catastrophic events.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Vinoth Seralan: Conceptualization (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Validation (equal);

Visualization (equal); Writing – review & editing (equal). **S. Leo Kingston:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Project administration (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Suresh Kumarasamy:** Conceptualization (supporting); Formal analysis (equal); Validation (supporting); Writing – review & editing (supporting). **Ajit Mahata:** Formal analysis (supporting); Visualization (supporting); Writing – review & editing (supporting). **Karthikeyan Rajagopal:** Validation (equal); Writing – review & editing (equal). **Tomasz Kapitaniak:** Conceptualization (equal); Validation (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article.

APPENDIX: DYNAMIC EVALUATION OF EXTREME EVENTS

In the main text, we mainly focused on the transition to extreme events via three different pathways and their relevant complexity measures. However, this [Appendix](#) illustrates the distinct dynamical evolution of extreme events to ensure their existence. The temporal dynamics of the chemical model and CO₂ laser that originated after the bounded chaotic motion are shown in [Figs. 8\(a\)](#) and [8\(c\)](#). The time series plots represent the frequent appearance of chaotic motion (dark to light green region) in a bounded range and

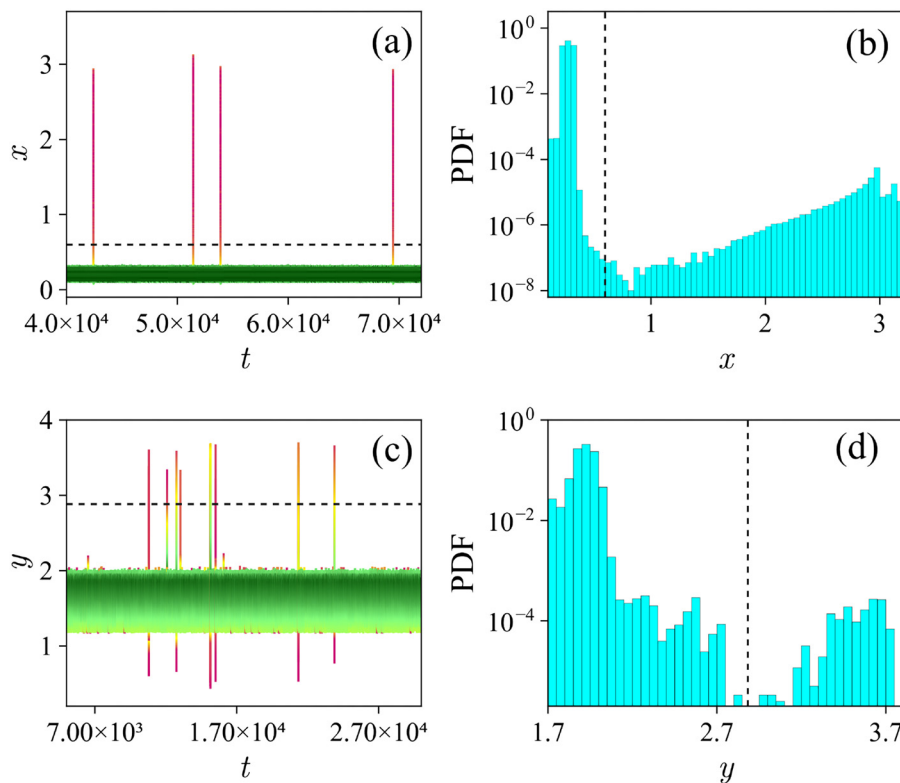


FIG. 8. Time evolution of extreme events: (a) Brusselator model and (c) CO₂ laser show rare existence of large-amplitude events from the bounded chaotic motion. Its corresponding PDFs show a heavy-tail distribution in (b) and (d). The horizontal and vertical dashed lines in (a)–(d) signify the qualifier threshold values (Q_{th}).

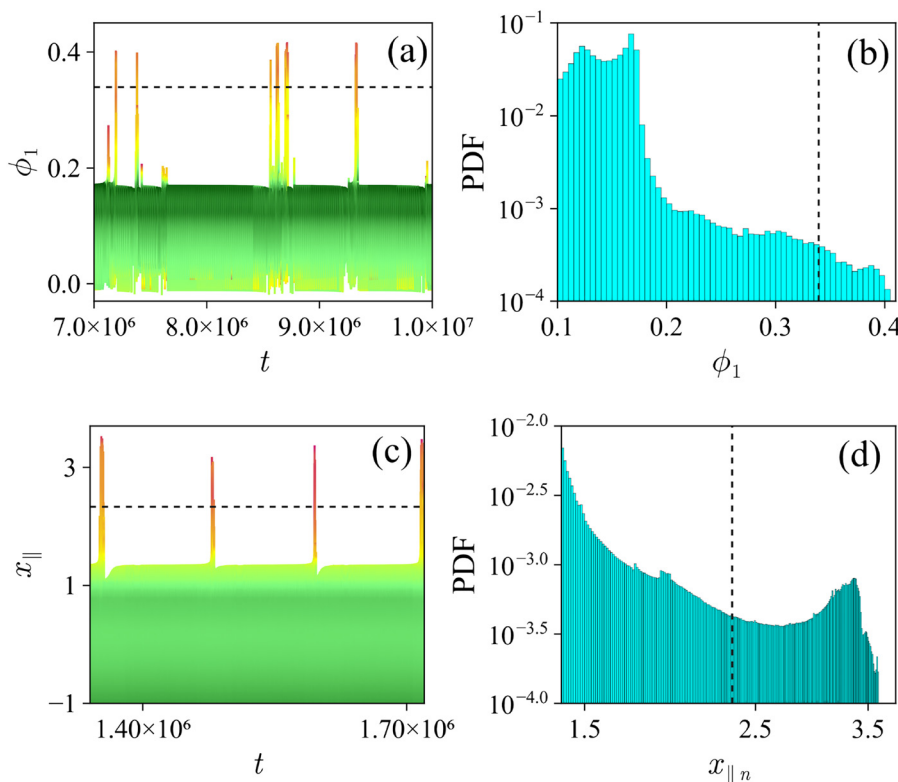


FIG. 9. Time series illustrating (a) quasiperiodic intermittency and (c) Pomeau–Manneville intermittency through extreme events. The rare bursting originated from the bounded quasiperiodic or periodic region that crossed the Q_{th} threshold values shown as horizontal dashed lines. The PDF of SQUID shows a heavy-tail distribution in (b), and the coupled Hindmarsh–Rose model exhibits a dragon-kings-like distribution (d).

occasional deviation of large-amplitude events (pink trajectories) at sporadic time intervals. The probability distribution functions (PDFs) of extreme events that appeared from the chaotic region are estimated for an extensive length of peaks of time series data portrayed in Figs. 8(b) and 8(d). Both systems show a heavy-tail distribution that confirms the existence of extreme events. The Q_{th} values are plotted as dashed lines, and the rare events exceeded the qualifier threshold values.

Moreover, we have presented the time series of extreme events, which originates from quasiperiodic intermittency and Pomeau–Manneville intermittency in Figs. 9(a) and 9(c). The unexpected flickering (pink trajectories) occurred from the quasiperiodic region (green) at infrequent times and is visible in Fig. 9(a). On the other hand, the formation of rare bursting from the periodic motion is portrayed in Fig. 9(c). Finally, we have computed the PDFs for the quasiperiodic intermittency and Pomeau–Manneville intermittency via large-amplitude dynamics, which are delineated in Figs. 9(b) and 9(d). Once again, the SQUID trimer manifests a heavy-tail distribution [see Fig. 9(b)]. However, the coupled Hindmarsh–Rose model exhibits a dragon-kings-like distribution, as illustrated in Fig. 9(d), indicating a significant deviation of rare events in the tails. As a consequence, there exists a hump-like structure in their probability distribution.

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